



Programming Fundamentals

G-LAB # 2

Course **Instructor Name:**

LAB Instructor:

Instructions:

- You have to submit CPP files of the program in a folder, named your Registration Number.
- You have to upload your lab tasks on CMS.
- All program codes should be written in C/C++. Students should use Visual studio compiler for coding.
- Indent and comment your code.
- Use meaningful variable names
- Plan your code carefully on a piece of paper before you implement it.
 - This lab focuses on using **character arrays only** (no string type).
 - You are given the `main()` function — **do not modify it.**
 - You must write **6 functions** based on the prototypes listed below.
 - Use **modular programming**: each function should handle **one specific task**.
 - Do not use built-in string functions (e.g., `strlen`, `strcpy`, `isalpha`, etc.).
 - Input will be read from a file named **input.txt**.

Learning Objectives:

- Demonstrate knowledge of basic **functions** programming C++.
-

Question 1:

Objective: Students will write functions to read a sentence from a file, count vowels, consonants, digits, and spaces using only character arrays (no strings).

Given `main()` Function (Students Cannot Modify):

```
#include<iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    char sentence[100];
```

```
    readFromFile(sentence);
```

```
    int vowels = countVowels(sentence);
```

```
    int consonants = countConsonants(sentence);
```

```
    int digits = countDigits(sentence);
```

```
int spaces = countSpaces(sentence);

displayStats(vowels, consonants, digits, spaces);

return 0;
}
```

Student Task: Write These Functions

1. `void readFromFile(char sentence[]);`
*Read a line from **input.txt** into a character array.*
2. `int countVowels(const char sentence[]);`
Count 'a', 'e', 'i', 'o', 'u' (both upper and lowercase).
3. `int countConsonants(const char sentence[]);`
Count all alphabetic characters except vowels.
4. `int countDigits(const char sentence[]);`
Count numeric digits '0' to '9'.
5. `int countSpaces(const char sentence[]);`
Count space characters ' '.
6. `void displayStats(int v, int c, int d, int s);`
Display all counts with proper labels.

Sample `input.txt`:

Hello Student! You have 2 tasks and 1 file.

Expected Output:

Vowels: 11

Consonants: 21

Digits: 2

Spaces: 8

End of Graded lab 2

